

When $r = 1$, $M(1, L, c_2)$ is isomorphic to $S^{[c_2]}$, the Hilbert scheme of 0-dimensional closed subschemes in S of length c_2 . Fogarty showed that $S^{[n]}$ ($n \in \mathbb{Z}_{\geq 0}$) is a smooth projective variety of dimension $2n$ [Fog68] and computed its Picard group [Fog73]. The Betti numbers of $S^{[n]}$ were completely determined by Göttsche [G90]. When $r \geq 2$, $M(r, L, c_2)$ can be reducible or non-reduced. But starting from Donaldson's generic smoothness result [Don90], the study of $M(r, L, c_2)$ has revealed that $M(r, L, c_2)$ has better geometric properties as we fix $r \geq 2$, L , and let c_2 increase. For example, when c_2 is large enough, Gieseker–Li [GL94] [GL96] and O'Grady [O'G96] showed that $M(r, L, c_2)$ is irreducible, normal, and generically smooth of expected dimension. Some asymptotic phenomena also occur in the Picard group and Betti numbers of $M(r, L, c_2)$ ($r \geq 2$) [Qin92][Li96][Li97][CW22].

This raises a natural question:

Question 1.1. *Are there asymptotic phenomena similar to those above in the geometry of $M(r, L, c_2)$ when $r = 0$?*

Sheaves on S of rank zero are supported on points or curves. In this paper, we focus on those pure sheaves supported on curves, which are called one-dimensional sheaves. In contrast to the positive rank case, there are few general results for the moduli space of one-dimensional sheaves. An essential difficulty is that the supports of one-dimensional sheaves can be singular, even non-reduced. To distinguish this case, we use the following notation. Given a nonzero effective divisor β on S and an integer χ , denote by $M_{\beta, \chi}$ the moduli space parametrizing polystable one-dimensional sheaves F on S , with determinant $\det(F) \cong \mathcal{O}_S(\beta)$ and Euler characteristic $\chi(F) = \chi$. We will make the following assumption on β , unless otherwise stated.

Assumption 1.2. *The following conditions hold:*

- (1) β is base-point-free.
- (2) A general curve in $|\beta|$ is smooth and connected.

A conjecture [SZ24, Conjecture 1.3] was proposed on the stabilization of (intersection cohomology) Betti numbers of $M_{\beta, \chi}$ when χ is fixed and β is sufficiently positive, as an analogue of [CW22, Conjecture 1.1]. When S is a del Pezzo surface, this conjecture was resolved in [PSSZ24] by proving an asymptotic formula of the refined BPS invariants which also refine the Betti numbers.

The primary motivation of this paper is to explore the stabilization of the Picard number $\rho(M_{\beta, \chi})$ of $M_{\beta, \chi}$ when β is sufficiently positive. Here, the Picard number $\rho(X)$ of a proper scheme X is defined as the dimension of the $\mathbb{Q}$ -vector space $\text{Num}_{\mathbb{Q}}(X)$ of the numerically equivalence classes of line bundles on X with rational coefficients, in the sense of [Kle66]. Fix a basis $[L_i]$ ($i = 1, \dots, \rho(S)$) for $\text{Num}_{\mathbb{Q}}(S)$, where each L_i is a smooth, connected effective divisor. We specify a positivity condition for β as follows.

Definition 1.3. *We say a divisor β on S satisfies (P) if*

- (1) β satisfies Assumption 1.2.
- (2) The arithmetic genus $p_a(\beta) > 0$.
- (3) For each $1 \leq i \leq \rho(S)$, there exists a smooth and connected curve $L'_i \in |\beta - L_i|$ such that L_i meets L'_i transversally with intersection number $L_i \cdot L'_i \geq |\chi| + 1$.

Note that for an ample divisor β , the multiple $n\beta$ satisfies (P) for all $n \gg 0$.

Our first main result is the following. We obtain an asymptotic lower bound of the Picard number of $M_{\beta, \chi}$.